



Figure 4: Effectiveness of our method.

Methods	R-1	R-5	R-10	mAP
ResNet50 baseline	76.9	87.8	91.0	58.7
ResNet50+ST	87.7	94.1	95.8	72.2
DenseNet121 baseline	79.3	89.9	92.6	63.3
DenseNet121+ST	90.8	95.2	96.5	76.9
PCB(*)	83.8	91.7	94.4	69.4
PCB+ST	94.0	97.0	97.8	82.8

Table 3: Generalization of the st-ReID on DukeMTMC-reID.

sub-module to see how the spatial-temporal stream makes an effect in the st-ReID. In this case, the st-ReID model is degraded as the PCB model. As shown in 4 (a), we can see that without the spatial-temporal probability estimation stream, the rank-1 accuracy drops 10.2% (from 94.0% to 83.8%).

Effectiveness of the joint metric. To show the effectiveness of the joint metric, we set a baseline by using Eqn. 5. In the baseline, both the VIS stream and the ST stream are normalized. For the fair comparison, we use the same VIS and ST streams. As shown in Figure 4 (b), our joint metric method improves the performance from 86.9% to 94.0%. Compared with the VIS stream (PCB model), the baseline also obtains a 3.1% improvement because it integrates the spatial-temporal information.

Influence of parameters. To investigate the impact of two important parameters in our st-ReID, i.e., the smoothing factor λ and the shrinking factor γ , we conduct two sensitivity analysis experiments. As shown in Figure 4 (c) and (d), when λ is in the range of 0.4~2.8 or γ is in the range of 1~7, our model nearly keeps the best performance.

Generalization of the st-ReID. To show the good generalization of the st-ReID, we further use different deep models as the VIS streams, respectively. The deep models are ResNet-50 (a clear model with the cross entropy loss), DenseNet-121 (a clear model with the cross entropy loss) and PCB. As shown in Table 3, it is observed that when adding the ST stream into these VIS streams and using our joint metric, we can achieve more than 10% improvement.

Conclusion

In this paper, we propose a novel two-stream spatial-temporal person ReID (st-ReID) framework that mines both the visual semantic similarity and the spatial-temporal information. Without bells and whistles, our st-ReID method achieves rank-1 accuracy of 98.1% on Market-1501 and 94.4% on DukeMTMC-reID, improving from the baseline 91.2% and 83.8%, respectively, outperforming all previous state-of-the-art methods by a large margin.

We intend to extend this work in two directions. First, the st-ReID builds a bridge between the conventional ReID and the cross-camera multiple object tracking and thus can be easily generalized to the cross-camera multiple object tracking. Second, we intend to further improve the performance of the st-ReID method using an end-to-end training manner.

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